Name: ______________________

Class: ______________________

Mr. Rawson • GCSE Computer Science Year 11 2026

Programming 2 — The Three Constructs

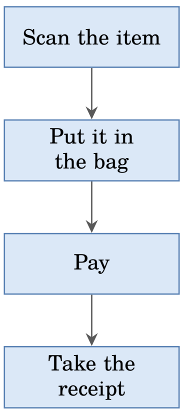
Sequence • Selection • Iteration — one machine, three ways — OCR J277 §2.2.1

Three things. That is the whole list.

Every program ever written — your greeting program, Minecraft, the software flying an aircraft — is built from **three ways of controlling what happens next**. There is no fourth.

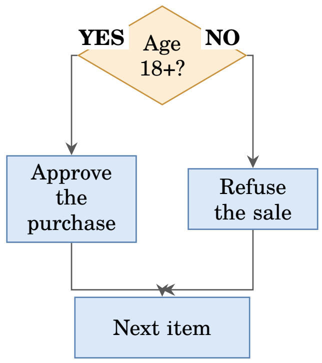
OCR asks for them by name (§2.2.1), so learn the words as well as the ideas. On this sheet all three run the same machine: **a supermarket self-checkout**.

1 SEQUENCE



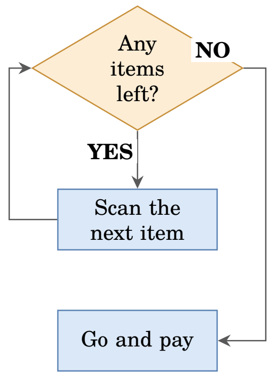
*Do things **in order**, one after another.*

2 SELECTION



***Choose** a path. Only one branch runs.*

3 ITERATION



***Repeat** until something changes.*

VOCAB Sequence (OCR §2.2.1) — instructions carried out in the order they are written, one after another

VOCAB Selection (OCR §2.2.1) — choosing between two or more paths, based on a condition

VOCAB Iteration (OCR §2.2.1) — repeating a set of instructions — also called a loop

Look at the shapes, not the words. A **rectangle** always does something. A **diamond** always asks a question — and it is the only shape with more than one arrow leaving it. **An arrow that goes back up the page is a loop.** Those three facts let you read any flowchart in the exam.

The four shapes — OCR uses these, so you must know them

Terminal — Where a program begins or ends

Process — Do something

Input / Output — Anything in or out

Decision — Asks a question

Name the construct

Write **Sequence**, **Selection** or **Iteration** beside each one. Some are harder than they look.

What the program does	Which construct?
Asks your name, then asks your age, then says hello	
Keeps asking for a password until you get it right	
Prints "Well done" only if your score is over 50	
Prints the 5 times table, from 5 up to 60	
Charges half price if you are under 16, full price otherwise	
Turns the heating on, waits an hour, turns it off	

The one people get wrong

"Keeps asking until you get it right" is *Iteration*, not Selection — even though it obviously involves a question.

The test is not *is there a condition?* Both of them have conditions. The test is: **does anything happen more than once?** If the same instruction can run twice, it is iteration. **A diamond on its own is selection; a diamond with an arrow going back up is iteration.**

3 examples of these terms using code you already know

You have already used all three. Here they are with their real names.

1 Sequence — no keyword at all. It is simply what Python does if you do not tell it otherwise: top line first, bottom line last.

```
item = input("Scan an item: ")
print("Added: " + item)
print("Please pay.")
```

Swap two of those lines and the program still runs — but it is now wrong. ***Order is a decision, even when you never typed a keyword.***

2 Selection — `if`, `elif`, `else`.

```
age = int(input("Your age? "))

if age >= 18:
    print("Sold.")
else:
    print("Calling an assistant.")
```

***Exactly one** of those two lines prints. Never both, never neither.* `elif` *adds a third path — and it is **one word**.* `else if` *is a syntax error in Python.*

3 Iteration — `for` when you know how many times, `while` when you do not.

```
for i in range(3):
    print("Scanning...")

while basket != "empty":
    print("Scanning...")
```

OCR names these two kinds and may ask for the difference. **`for`** ***is count-controlled** — you decide the number in advance.* **`while`** ***is condition-controlled** — it keeps going until something changes, and it might never stop.*

> **VOCAB** **Count-controlled loop** (OCR §2.2.1) — repeats a fixed number of times, decided in advance — a `for` loop

> **VOCAB** **Condition-controlled loop** (OCR §2.2.1) — repeats until a condition changes — a `while` loop

You have already written all three

Look back at Programming 1

The challenge program you wrote by hand — name, number of greetings, greet that many times — contains **all three constructs**, and you were not told that at the time.

```
name = input("What is your name? ")
times = int(input("How many times? "))
for i in range(times):
    print("Hello, " + name)
```

Find them. Which lines are sequence? Which line starts the iteration? And — harder — **where is the selection?**

(Careful. That one is a trick, and the answer is worth understanding rather than guessing.)

Build it — the self-checkout

Write a program that uses **all three constructs**. You have everything you need.

1. Ask how many items are in the basket. *(sequence)*
2. Scan that many items, asking for the price of each. *(iteration)*
3. Add each price to a running total.
4. At the end, if the total is over £50, print "Free delivery!" — otherwise print the delivery charge. *(selection)*

Now try • Get steps 1 and 2 working first. Print the prices back out to check.
Now try • Add the running total. *Start it at `0` before the loop, not inside it* — try it both ways and see what goes wrong.
Now try • Add the selection at the end.
Now try • **Stretch:** draw your finished program as a flowchart, using the four shapes.

Now write it out — by hand

main.py

```

```

Save before you close the tab. Paper 2 is written on paper.